

Midterm

Time: Apr. 1 Tuesday, in class (1:00-2:40pm)

Location: 教学中心101

Covers Chapter 2~7

Format: 10 true-false questions and 4 problems; closed-book, one A4-size cheat sheet allowed

Grade: %35 of the total grade

Midterm review

Midterm exam

Time

- 13:00 – 14:40 pm, April 1, 2025.

Location

- 教学中心101
- Seat arrangement TBA

Format

- Closed-book. You can bring an A4-size cheat sheet and nothing else
- 10 true-false questions and 4 problems

Grade

- 35% of the total grade
- 计算器 ❌
- 涂卡笔 ✅

Midterm exam

Disclaimer

- Topics covered in this review may not appear in the exam.
- Topics not covered in this review may appear in the exam.

Algorithm analysis

Algorithm analysis

- Comparison of algorithm complexity
 - Time, memory
- Asymptotic order of growth
 - O , o , Ω , ω , Θ
 - Properties of the complexity operation
- Analyzing complexity
 - Loops, if-else, recursion
- Several ways to solve a recurrence relation
 - Guess, substitution, recursion tree, master method

Divide-and-Conquer

Divide-and-conquer

- Basic idea
- Analysis of D&C complexity
 - Finding recurrence relation
 - Solve the recurrence relation
- Algorithms
 - Merge-sort
 - Closest Pair of Points
 - Integer Multiplication
 - Matrix Multiplication
 - Convolution and FFT

Greedy algorithm

Greedy algorithms

- Basic idea
- The selection measurement of (local) optimality
- Analysis of greedy algorithm complexity
 - Greedy algorithm stays ahead
 - Exchange argument
- Algorithms
 - Interval Scheduling: Choose the job with the earliest finish time
 - Scheduling to Minimize Lateness: Choose the job with the earliest deadline
 - Clustering: Single-link k-clustering
 - Optimal Caching: Evict item that is requested farthest in future

Dynamic programming

Dynamic programming algorithms

- Comparison to greedy and divide-and-conquer algorithms
- Basic steps
 - Define the sub-problems: $OPT(\dots)$
 - Write down the recursive formulas, Ex: $OPT(i) = \max(f(OPT(j)), g(OPT(k)), \dots)$, $j, k < i$
 - Compute the formulas either bottom-up or top-down
- Analysis of dynamic programming algorithm complexity
- Algorithms
 - Weighted interval scheduling: 1D array; binary choice
 - Knapsack: 2D array; adding a new variable (weight limit)
 - RNA secondary structure: 2D array; intervals
 - Sequence Alignment: 2D array; prefix alignment
 - Sequence Alignment in Linear Space: Combination of divide-and-conquer and dynamic programming
 - Shortest path with negative edges: (Bellman-Ford) 2D array; shortest path with edge number $\leq i$

Network flow

Network flow problem

- Concepts: s-t flow, max-flow, s-t cut, min-cut
- Max-flow min-cut theorem
- Ford-Fulkerson Algorithm
 - Initialize with empty flow
 - Find an augmenting path P in the residual graph G_f
 - Augment flow along path P and update G_f
 - Repeat until you get stuck
- Application of network flow problem
 - Bipartite Matching
 - Disjoint Paths
 - Circulation with Demands
 - Survey Design
 - Image Segmentation
 - Project Selection
 - Baseball Elimination

Good luck!